HW #2

1.

a) $x(n)=\left(\frac{1}{2}\right)^n u(n)$, $h(n)=\delta(n-1)+\left(\frac{1}{2}\right)^n u(n)$

$$y(n)=x(n)\otimes h(n)=\sum_{k=-\infty}^{\infty}x(k)h(n-k)=\sum_{k=-\infty}^{\infty}\left(\frac{1}{2}\right)^k u(k)\left[\delta(n-k-1)+\left(\frac{1}{2}\right)^{n-k}u(n-k)\right]$$

$$=\sum_k\left(\frac{1}{2}\right)^k u(k)\delta(n-k-1)+\sum_k\left(\frac{1}{2}\right)^k u(k)\left(\frac{1}{2}\right)^{n-k}u(n-k)$$

$$=\underbrace{\left(\frac{1}{2}\right)^{n-1}u(n-1)}_{n\geq 1}+\underbrace{\sum_{k=0}^{n}\left(\frac{1}{2}\right)^k\left(\frac{1}{2}\right)^{n-k}}_{z(n)}$$

$n-k\geq 0 \Rightarrow k\leq n$, $k\geq 0$, $0\leq k\leq n$

$$z(n)=\sum_{k=0}^{n}\left(\frac{1}{2}\right)^k\left(\frac{1}{2}\right)^{n-k}=\sum_{k=0}^{n}\left(\frac{1}{2}\right)^n=\left(\frac{1}{2}\right)^n(n+1)$$

$$\Rightarrow\quad y(n)=\left(\frac{1}{2}\right)^{n-1}u(n-1)+\left(\frac{1}{2}\right)^n(n+1)u(n)$$

b).

$x(n)=nu(n)$, $h(n)=u(n)-u(n-10)$

$$y(n)=x(n)\otimes h(n)=\sum_{k=-\infty}^{\infty}x(k)h(n-k)=\sum_{k=-\infty}^{\infty}k\,u(k)\,h(n-k)$$

$k\geq 0$ $\quad 0\leq n-k\leq 9$

$-9\leq k-n\leq 0$

$n-9\leq k\leq n$

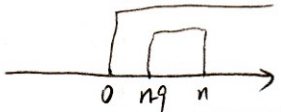
① $n<0$ $\quad y(n)=0$

② $n-9<0 \Rightarrow n<9$, $n\geq 0$ $\Rightarrow 0\leq n<9$



$$y(n)=\sum_{k=0}^{n}k=\frac{n(n+1)}{2}$$

③ $n-9\geq 0 \Rightarrow n\geq 9$



$$y(n)=\sum_{k=n-9}^{n}k=\frac{10(n+n-9)}{2}=5(2n-9)$$

$\Leftarrow$ $\sum_{n=a_1}^{a_2}k=\frac{(a_2-a_1+1)(a_2+a_1)}{2}$

$$y(n)=\begin{cases}0 & n<0\\ \frac{n(n+1)}{2} & 0\leq n<9\\ 5(2n-9) & n\geq 9\end{cases}$$

(c) $x(n) = [1, -2, 4, -8, 16]$, $h(n) = [1, -1, 1, -1]$

$y(n) = \sum_{k=-\infty}^{\infty} x(k)\,h(n-k)$

$h(-n) = [-1, 1, -1, 1]$

$x(k)$			1	-2	4	-8	16	
$h(-2-k)$	-1	1	-1	1				$y(-2) = 1$
$h(-1-k)$		-1	1	-1	1			$y(-1) = 1\times(-1) + (-2)\times 1 = -3$
$h(0-k)$			-1	1	-1	1		$y(0) = 1\times 1 + (-2)\times(-1) + 4\times 1 = 7$
$h(1-k)$				-1	1	-1	1	$y(1) = 1\times(-1) + (-2)\times 1 + 4\times(-1) + (-8)\times 1 = -15$
$h(2-k)$					-1	1	-1	1 $y(2) = (-2)\times(-1) + 4\times 1 + (-8)\times(-1) + 16\times 1 = 30$
$h(3-k)$						-1	1	-1 1 $y(3) = 4\times(-1) + (-8)\times 1 + 16\times(-1) = -28$
$h(4-k)$							-1	1 -1 1 $y(4) = (-8)\times(-1) + 16\times 1 = 24$
$h(5-k)$								-1 1 -1 1 $y(5) = 16\times(-1) = -16$

$y(n) = [1, -3, 7, -15, 30, -28, 24, -16]$

(d) $x(n) = [1, 2, 3, 0, -1]$ $h(n) = [2, -1, 3, 1, -2]$

$h(-n) = [-2, 1, 3, -1, 2]$

$x(k)$			1	2	3	0	-1		
$h(-3-k)$	-2	1	3	-1	2				$y(-3) = 1\times 2 = 2$
$h(-2-k)$		-2	1	3	-1	2			$y(-2) = 1\times(-1) + 2\times 2 = 3$
$h(-1-k)$			-2	1	3	-1	2		$y(-1) = 1\times 3 + 2\times(-1) + 3\times 2 = 7$
$h(0-k)$				-2	1	3	-1	2	$y(0) = 1\times 1 + 2\times 3 + 3\times(-1) + 0\times 2 = 4$
$h(1-k)$					-2	1	3	-1 2	$y(1) = 1\times(-2) + 2\times 1 + 3\times 3 + 0\times(-1) + (-1)\times 2 = 7$
$h(2-k)$						-2	1	3 -1 2	$y(2) = 2\times(-2) + 3\times 1 + 0\times 3 + (-1)\times(-1) = 0$
$h(3-k)$							-2	1 3 -1 2	$y(3) = 3\times(-2) + 0\times 1 + (-1)\times 3 = -9$
$h(4-k)$								-2 1 3 -1 2	$y(4) = 0\times(-2) + (-1)\times 1 = -1$
$h(5-k)$								-2 1 3 -1 2	$y(5) = (-1)\times(-2) = 2$

$y(n) = [2\ 3\ 7\ 4\ 7\ 0\ -9\ -1\ 2]$